

Yige Yan

Ph.D. Candidate, College of Computing and Data Science, Nanyang Technological University

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Research Profile

Research Interests: Contrast-free cardiac MRI; weakly supervised medical image segmentation; ageing and neurodegeneration; pathology–age disentanglement; functional connectivity; multimodal learning; model auditing and trustworthy medical AI.

Research Summary: I make already-acquired clinical data yield diagnostic information it was not designed to carry. In cardiac imaging I recover microvascular obstruction from contrast-free cine sequences, where the acquisition that defines the finding is absent. In neuroimaging I remove the age-related variation that dominates functional connectivity, where the disease signal is present but not separable. Both lines extend what existing scans can support, with no new acquisition protocol and no contrast agent. A parallel line audits whether a trained model has learned that evidence or a shortcut that scores the same.

Availability: Research internship in medical AI or computer vision, before thesis submission in the first half of 2027; postdoctoral positions from 2027.

Education

Nanyang Technological University

Singapore

Ph.D. in Computer Science and Engineering, GPA: 4.67/5.00

Aug 2023 – Present

Main supervisor: Prof. Jagath C. Rajapakse (IEEE Fellow), College of Computing and Data Science.

Co-supervisor: Assoc. Prof. Nagaendran Kandiah, Lee Kong Chian School of Medicine.

Expected thesis submission: first half of 2027.

Nanyang Technological University

Singapore

M.Sc. in Computer Control and Automation, GPA: 4.44/5.00

Aug 2022 – Jun 2023

Supervisor: Prof. Wee Peng Tay, School of Electrical and Electronic Engineering.

Hohai University

Nanjing, China

B.Eng. in Computer Science, GPA: 4.48/5.00

Sep 2017 – Jun 2021

Supervisor: Dr. Xiaoqin Zhou, College of Artificial Intelligence and Automation.

Publications & Manuscripts

Peer-reviewed

1. **Yige Yan**, Yi Hao Chan, Jun Cheng, Qian Hui Chew, Kang Sim, and Jagath C. Rajapakse. “Latent Pathology–Age Decoupling in Functional Connectivity Graphs for Multi-task Schizophrenia Assessment.” *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026. **Early Accept.**
2. **Yige Yan**, Jun Cheng, Xulei Yang, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Dual Correlation-Aware Mamba for Microvascular Obstruction Identification in Non-contrast Cine Cardiac Magnetic Resonance.” *MICCAI*, 2025. **Early Accept.**
3. **Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Coarse-Grained Mask Regularization for Microvascular Obstruction Identification from Non-contrast Cardiac Magnetic Resonance.” *MICCAI*, 2024. **Early Accept.**

Under review

4. **Yige Yan**, Jing Xia, Yi Hao Chan, Thomas Welton, and Jagath C. Rajapakse. “Decoupling Age Effects from Parkinson’s Disease Functional Connectivity.” *Medical Image Analysis*.
5. **Yige Yan**, et al. “When Should an Image Swap Change a Vision–Language Model’s Answer? Evaluating Answer Preservation and Correct Updating.” *International Conference on Learning Representations (ICLR)*, 2027.
6. **Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Spatiotemporal Adaptive Mamba with Coarse-Grained Mask Regularization

for Microvascular Obstruction Identification from Non-contrast Cine Cardiac MRI.” *Neurocomputing*.

In preparation

7. **Yige Yan** and Jagath C. Rajapakse, for the Alzheimer’s Disease Neuroimaging Initiative. “Clinical Evidence Reconfigures Age Reliance in Language Models Assessing Alzheimer’s Disease.” *Target: npj Digital Medicine*.
8. **Yige Yan**, et al. “Temporal Variation Proxy and Staged Proposal–Support Integration for Contrast-Free MVO Segmentation in Cine CMR.” *Target: IEEE Transactions on Medical Imaging*.
9. **Yige Yan**, et al. “Auditing Named Knowledge Interfaces: Regularization, Responsiveness, and Predictive Utility.” *Target: Knowledge-Based Systems*.

Conference Presentations & Posters

- *MICCAI 2026*, Strasbourg, France, Sep–Oct 2026. Poster presentation.
- *MICCAI 2025*, Daejeon, Republic of Korea, Sep 2025. Poster presentation.
- *IEEE EMBS – Duke-NUS – HealthTEC.SG – NUS iHealthtech Forum 2025: AI in Healthcare and MedTech*, Singapore, Apr 2025. Poster presentation.
- *MICCAI 2024*, Marrakesh, Morocco, Oct 2024. Poster presentation.
- *Singapore Healthcare Engineering Conference (SHEC) 2024*, IEEE EMB Singapore Chapter, Singapore, Oct 2024. Poster presentation.
- *AI Health Summit 2023*, SingHealth, Ministry of Health and A*STAR, Singapore, Nov 2023. Poster presentation.

Research Experience

Nanyang Technological University

Ph.D. Researcher, College of Computing and Data Science

Singapore

Aug 2023 – Present

- **Contrast-free cardiac MRI.** The sequence that defines microvascular obstruction requires gadolinium and is absent from routine cine acquisition. Built a multi-year line recovering the finding from cine alone, from coarse-grained supervision under imperfect LGE-to-cine alignment through motion modeling and spatiotemporal state-space learning to pixel-level segmentation.
- **Confounder-free functional connectivity.** Disease signal is present but dominated by age-associated variation. Designed graph-based representation learning that disentangles the two and reintroduces age as controlled clinical context, across schizophrenia, Parkinson’s disease and Alzheimer’s disease.

Institute for Infocomm Research (I²R), A*STAR

Research Student Attachment, Medical Imaging

Singapore

Aug 2023 – Present

- Microvascular obstruction identification from non-contrast cine cardiac MRI, with collaborators across A*STAR, NTU, Duke-NUS Medical School and the National Heart Centre Singapore.

Institute for Infocomm Research (I²R), A*STAR

Research Student Attachment, OCTA Denoising

Singapore

Mar 2023 – Jun 2023

- Deep-learning denoising and enhancement for optical coherence tomography angiography (OCTA), with collaborators across A*STAR, NTU and the Singapore National Eye Centre.

Research Grants & Proposal Experience

Singapore MOE Academic Research Fund (AcRF) Tier 2

Primary proposal writer and research-plan developer

In preparation, Sep 2026

Faculty PI: Prof. Jagath C. Rajapakse

“Separating and Contextualizing Age for Reliable Assessment and Prognosis of Neurodegenerative Disease.”
Drafted the research plan and preliminary-study design, extending my doctoral research on pathology–age

decoupling into a multi-project research program.

Teaching Experience

Nanyang Technological University

Singapore

Teaching Assistant, College of Computing and Data Science

2024 – Present

- **SC4001 Neural Networks and Deep Learning**, lead teaching assistant. Coordinated the teaching-assistant group and authored assignment problems. Jan–Apr 2025, Aug–Nov 2025, Aug 2026–present.
- **SC1013 Physics for Computing**, lab supervisor. Aug–Nov 2025.
- **SC1006 Computer Organisation & Architecture**, lab supervisor. Feb–Mar 2025.
- **SC2079 Multidisciplinary Design Project**, lab supervisor. Aug–Oct 2024.
- **SC2008 Computer Network**, lab supervisor. Feb–Apr 2024.

Research Mentoring

Nanyang Technological University

Singapore

Mentor, Undergraduate Final-Year Projects (FYP)

2025 – Present

- AY2026/27. “Predicting Multiple Cognitive Functions Simultaneously from Brain Functional Connectivity via an Interpretable Multi-Task Learning Network.”
- AY2026/27. Age-conditioned Riemannian residual modeling of schizophrenia functional connectomes.
- AY2025/26. “Explainable AI for Functional MRI Deep Learning Model Analysis.”
- AY2025/26. “Multi-Modal Functional MRI Fusion for Fluid Intelligence Prediction.”
- AY2025/26. Graph state-space modeling for Alzheimer’s disease neuroimaging.

Academic Service

Reviewer: MICCAI; IEEE Transactions on Neural Networks and Learning Systems (TNNLS); IEEE Transactions on Image Processing (TIP).

Honors, Awards & Scholarships

- **MICCAI 2024 Registration and Travel Award.**
- **Nanyang Technological University Research Scholarship (RSS).**
- **Outstanding Undergraduate Graduation Project**, Hohai University, 2021.
- **Outstanding Student Award**, Hohai University.
- **Academic Merit Scholarship**, Hohai University.

Technical Skills

Programming & Frameworks: Python, PyTorch, PyTorch Geometric, C++, C, NumPy, SciPy, scikit-learn.

Methods: Deep learning, graph neural networks, state-space models (Mamba), medical image analysis, multimodal learning, statistical evaluation.

Domains: Medical image analysis, cardiac MRI, fMRI, neuroimaging, OCTA, medical multimodal AI.

Languages: Chinese (native), English (fluent), Japanese (proficient).